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A causal attention network with time frequency channel feature fusion for epileptic seizure prediction

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ABSTRACT

Background: Epilepsy poses ongoing physical and mental threats and causes substantial economic burdens. Better seizure forecasting enables faster medical responses, improving patients' quality of life and lowering healthcare costs. Research mainly focuses on early forecasting within a short preictal window, often too brief for effective drug administration. A major challenge is that a longer preictal phase may resemble the interictal state, making differentiation difficult.

New methods: We propose a causal attention network (CANet) with a longer interictal and preictal of 1 h and 2 h respectively as the research object. In the feature extraction, a dilated causal convolution network is employed to extract local features. Causal attention is innovatively incorporated into epilepsy prediction to capture global correlation features. The complementary integration of these two methods enhances feature extraction and enables a more precise distinction between interictal and preictal periods. A double-layer dynamic window algorithm is developed for seizure prediction.

Results: We evaluate the performance on Freiburg and CHB-MIT datasets. On the Freiburg dataset, the sensitivity(Sen) of the 1/2-hour preictal intervals was 100.00%/96.67%, with a false alarm rate per hour (FAR) of 0.0077/h/0.0472/h, and the average prediction time (APT) was 97.59 min. On the CHB-MIT dataset, we achieved Sen of 97.06%/92.31%, FAR of 0.0251/h/0.0666/h, and APT of 94.85 min, under the same conditions.

Comparison with existing methods and conclusion: Our approach outperforms most of the previous methods, and the intracranial EEG (Freiburg) can more effectively distinguish interictal and preictal periods than scalp EEG (CHB-MIT).

1. Introduction

Epilepsy is a long-term neurological condition resulting from abnormal electrical impulses in brain cells. It is defined by abrupt and transient episodes. Its global prevalence ranges from 0.6% to 1%, with a notable proportion unresponsive to antiepileptic medications (Dalic and Cook, 2016; Kuhlmann et al., 2018). The condition lead to cognitive decline, psychological issues, and significant economic burden due to frequent medical visits and long-term medication use (Fisher et al., 2000). Therefore, precise and prompt forecasting of epileptic seizures has received extensive attention and research worldwide.

Administering antiepileptic medications promptly during the preictal stage can significantly reduce the likelihood of seizure occurrence (Assali et al., 2023; Litt and Echauz, 2002; Tetzlaff and Senger, 2012). Rescue medications must achieve onset of action within 5 to 30 min to prevent progression to established status epilepticus; prolonged status epilepticus exceeding 30 min may result in cerebral damage (Trinka et al., 2015; Glauser et al., 2016). Fast-acting antiepileptic drugs work within 30 min and are mainly benzodiazepines, which pose key risks of respiratory and circulatory depression. They should only be used in facilities with resuscitation capabilities (Brophy et al., 2012). This underscoring the importance of maintenance antiepileptic drugs. Most Maintenance antiepileptic drugs take more than 30 min to take effect, and some even need over an hour (Johannessen Landmark et al., 2020; Meirinho et al., 2021; Patsalos et al., 2008). Recent research primarily centers around short-term epilepsy forecasting. The Period

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preceding seizure onset (SOP) is typically defined as half an hour or one hour, while the APT generally falls below half an hour (Yuan et al., 2020; Assali et al., 2023; Pinto et al., 2021). A sufficiently long APT is necessary to enable patients and caregivers to implement preventive actions promptly (Li et al., 2021; Gao et al., 2023; Guo et al., 2023).

Recent research findings a longer SOP and an extended APT suggest promising prospects of providing epilepsy patients with more time for proactive intervention (Trinka et al., 2015; Bruno et al., 2020; Hu et al., 2020). An alarm meter were created that can provide a warning period ranging from 5 to 960 min (with an average of 114 ± 151 min) (Cook et al., 2013). An approach utilized multi-day cycles captured through wearable technology were proposed to forecasting epileptic seizures (Xiong et al., 2023). These studies employed various methods to predict epileptic seizures within hours or days, and the experimental outcomes confirmed that it is possible to forecast epileptic seizures across extended periods. However, the challenge of accurately distinguishing between longer pre-seizure and inter-seizure phases has led to a reduction in the accuracy of epilepsy prediction research.

Currently, deep learning algorithms have achieved notable effectiveness in multiple domains, such as speech recognition and computer vision (Yadav et al., 2022; Long et al., 2022), and are increasingly being utilized in epileptic prediction. Several studies have used time–frequency analysis. Truong et al. (2018) combined short-time Fourier transformation (STFT) with convolution neural networks, while Assali et al. (2023) merged spectral and temporal features. In spatio-temporal modeling, Zhang et al. (2023) introduced the STCNN, and Liu et al. (2024) developed a 3D convolution-bidirectional LSTM-attention model. Regarding attention mechanisms, Bhattacharya et al. (2022) applied deep Transformers to time–frequency data, and Sun et al. (2021) proposed the CADCNN framework. These innovations have improved seizure prediction and advanced epilepsy forecasting.

Among the aforementioned methods, the maximum SOP of 60 min limits the APT nearly impossible to extend further. An extended APT period enables patients and caregivers to have sufficient time to intervene proactively in advance. When the APT is longer than that of conventional anti-epileptic drugs, taking medication after receiving a warning can effectively suppress seizures. Precise prediction alerts are expected to reduce the frequency of medication and dependence for those who require daily administration of medication at fixed intervals. However, in practice, extending the SOP still faces challenges. As the preictal duration increases, its electrophysiological characteristics begin to overlap with those of the interictal period. This similarity makes it progressively more challenging to differentiate preictal from interictal phases using feature-based analysis. For spectrograms generated by STFT, EEG signals are correlated both across time and between frequency domains. Additionally, each channel has its own unique characteristics. In previous studies, causal relationships at both the time–frequency and channel levels have rarely been considered.

Based on the above problems, we propose the causal attention network (CANet). This approach fully leverages the local feature extraction and fusion capabilities of convolution networks across various dimensions, along with the global causal feature extraction advantages of CANet. Initially, dilated causal convolution networks capture local causal features in the time and frequency dimensions, followed by convolution to fuse all channel features. Subsequently, a causal attention network is introduced firstly in epilepsy prediction research. It learns the global causal relationships within the data fed into the module. Additionally, we present a new double-layer dynamic window prediction algorithm, developed to minimize false positives and improve the timing of medication administration for better patient safety. The Freiburg dataset and the CHB-MIT dataset were used to evaluate our method, exhibiting better performance than many of the leading existing approaches.

The key contributions of our study are presented as follows:

- This is the first time introduces the Causal Attention Network to epilepsy seizure prediction. By building a global causal reasoning framework, the model captures deep-level correlations across EEG signals.
- The dilated causal Convolution Network is used to simultaneously extract the time-domain features and spectral domain features, and cross-channel feature fusion eliminates noise interference from a single channel.
- A double-layer dynamic window is used to evaluate false alarm errors and false positive rates, providing precise early warnings for patients.

2. Materials and methods

2.1. Data description

We evaluated our proposed CANet through comprehensive experiments on two well-known EEG datasets: Freiburg and CHB-MIT. The Freiburg dataset includes 87 seizures from 21 epilepsy patients (13 females, 8 males), aged 10–50 years. Each patient experienced 2 to 5 seizures. Recordings were made with 256 Hz sampling rate, 128 channels. Seizure initiation and epileptic-like patterns were labeled by experts (Aschenbrenner-Scheibe et al., 2003). The CHB-MIT database includes EEG recordings from 22 children: 17 girls (ages 1.5–19) and 5 boys (ages 3–22). Of these, chb21 and chb01 were recorded from the same female participant, separated by 1.5 years. All of whom had taken antiepileptic drugs for several days. To ensure consistency in the method and to maximize the utilization of data from more patients, we followed (Zhang et al., 2019; Park et al., 2018), collected the data using 18 standard channels at a sampling rate of 256 Hz (Anugraha et al., 2017).

Complex, irregular brain activity appears about 7 h before seizure onset, while electrophysiological patterns resembling typical epileptic seizures emerge approximately 2 h prior, signaling an impending seizure (Litt et al., 2001). To evaluate our method's performance and explore longer SOP prediction, we defined multiple 1-hour and 2-hour preictal segments per patient in both datasets. Consistent with Yang et al. (2021), the inter-seizure interval was defined as the period starting 4 h post-seizure and ending 4 h pre-next-seizure. Given the inherent imbalance favoring interictal data, we balanced the count of preictal and interictal segments per patient. Specifically, for each preictal period, we randomly selected an interictal segment of the same duration, repeating this process three times to incorporate diverse interictal data and ensure consistent segment counts between the training and testing phases (Wang et al., 2021). To fulfill the basic criteria for training and testing in experiments involving consistent 1-hour preictal intervals per patient, a minimum of two 1-hour preictal segments and two 1-hour interictal segments are required. Since some patients did not have at least two 1-hour preictal and interictal data periods, these patients were excluded from our study. Similarly, for experiments that uniformly used 2-hour preictal periods, the patient selection criteria were the same as those used for experiments with 1-hour preictal periods. Based on the aforementioned definitions and constraints, Tables 1 and 2 respectively outline the details of the qualified patients in the two datasets.

2.2. Pre-processing

Due to the inherent non-stationarity and high variability of raw EEG, time–frequency analysis methods are often used for preprocessing and feature extraction for epileptic seizure prediction (Usman et al., 2020; Peng et al., 2022). In this paper, STFT is used because it can provide local spectral representation and reveal the instantaneous changes of frequency components before epileptic seizures, which is difficult to obtain from time domain signals or a single global Fourier transform. We set the EEG window length to 30 s, following Truong et al. (2018). The core idea of the STFT is to apply a fixed-length window function to

Table 1
Details of qualified patients in the Freiburg dataset.

Patient ID	Sex (F/M)	Age	1h		2h	
			No.	Preictal(h)	No.	Preictal(h)
Burg01	F	15	3	6	2	4
Burg03	M	14	4	8	–	–
Burg04	F	26	4	8	2	4
Burg05	F	16	2	4	2	4
Burg06	F	31	3	6	2	4
Burg07	F	42	3	6	–	–
Burg08	F	32	2	4	–	–
Burg09	M	44	5	10	2	4
Burg10	M	47	5	10	2	4
Burg11	F	10	4	8	2	4
Burg12	F	42	4	8	2	4
Burg13	F	22	2	4	–	–
Burg14	F	41	3	6	–	–
Burg15	M	31	3	6	3	6
Burg16	F	50	5	10	2	4
Burg17	M	28	5	10	4	8
Burg18	F	25	5	10	4	8
Burg19	F	28	4	8	2	4
Burg20	M	33	5	10	4	8
Burg21	M	13	4	8	2	4
Totals	–	–	**75**	**150**	**37**	**74**

Table 2
Details of qualified patients in the CHB-MIT dataset.

Patient ID	Sex (F/M)	Age	1h		2h	
			No.	Preictal(h)	No.	Preictal(h)
Chb01	F	11	5	10	5	10
Chb02	M	11	2	4	2	4
Chb03	F	14	4	8	–	–
Chb04	M	22	3	6	3	6
Chb05	F	7	5	10	4	8
Chb06	F	1.5	9	18	6	12
Chb07	F	14.5	3	6	3	6
Chb09	F	10	4	8	3	6
Chb10	M	3	6	12	6	12
Chb13	F	3	3	6	3	6
Chb16	F	7	3	6	2	4
Chb18	F	18	3	6	2	4
Chb19	F	19	3	6	–	–
Chb20	F	6	3	6	2	4
Chb21	F	13	3	6	–	–
Chb22	F	9	3	6	3	6
Chb23	F	6	3	6	–	–
Totals	–	–	**65**	**130**	**44**	**88**

analyze a signal in the time domain. Captured segments are analyzed using a Fourier transform to obtain a representation of the local spectrum within a narrow time window around time t. The STFT is used to generate a series of local spectra for each time interval. As the window moves along the entire time axis, the STFT generates a series of localized spectra for each time interval. Consequently, STFT provides a 2-D time–frequency representation, and its core computational formulation is expressed as follows:

$$\mathrm{STFT}\{x(t)\}(t,\omega) = \int_{-\infty}^{\infty} x(\tau)\, f(\tau - t)\, e^{-j\omega\tau}\, d\tau \tag{1}$$

Here, $x(t)$ is the time-domain signal, t and τ denote the time, w denotes the frequency and $f(\tau - t)$ is a window function whose center is at time t, e^{-jwt} denotes the Fourier kernel. The output $\mathrm{STFT}\{x(t)\}(t,w)$ is a complex matrix representing the amplitude and phase of the EEG signal at each time–frequency point (Ozdemir et al., 2021). In this study, a series of multi-channel EEG signal data were input into the STFT using a sliding window approach. In order to show the specific input and output data changes more clearly, we uniformly denote "$C1$" as the number of 6 channels in the Freiburg dataset, "$C2$" as the number of 18 channels in the CHB-MIT dataset, "F" as the frequency value of 61, "T" as the time frame of 114, "T_S" as the sample size of 7680 in the time domain and "TF" as the number of time–frequency features in the STFT output of 6954. For the Freiburg dataset, the data input format was $C1 \times T_S$, representing 6 electrode channels, with each channel containing 7680-time samples. The STFT processes each channel independently. For each channel, the STFT decomposes the time series into 61 frequency bins and 114-time frames, thereby generating a frequency-time matrix of size $F \times T$ for each channel. Therefore, the overall output of the STFT across all channels forms a $C1 \times F \times T$ three-dimensional array, fully encapsulating the spectral information of the original signal as it evolves over time. In the CHB-MIT dataset, the only difference from the Freiburg dataset is the number of electrode channels is 18. This transformation enables us to track epileptic seizure activity across both the time and frequency domains, providing critical features for subsequent analysis.

The EEG recordings in the Freiburg and CHB-MIT datasets were exposed to 50 Hz and 60 Hz power line noise contamination, respectively (Truong et al., 2018, 2019). Power line noise can be effectively removed by filtering out components within specific frequency bands (Park et al., 2011). We use a bandpass filter to remove the current noise in the corresponding frequency band, enabling the model to learn more effective features. For the Freiburg dataset, we exclude components within the 47–53 Hz and 97–103 Hz bands to remove 50 Hz noise. Similarly, for the CHB-MIT dataset, we filtered out components in the 57–63 Hz and 117–123 Hz bands to eliminate 60 Hz interference. In addition, the DC component (0 Hz) was removed from all records (Ma et al., 2019). Due to the removal of specific frequency bands, other bands were naturally spliced together to reconstruct EEG signals with better data quality.

We saved these preprocessed spectrogram data as a NumPy matrix and used it as the input for module B (causal attention network with time frequency channel feature fusion) as shown in Fig. 1. The spectrogram of the Freiburg and CHB-MIT datasets are represented as NumPy matrices with dimensions $C1 \times F \times T$ and $C2 \times F \times T$, respectively. Here, $C1$ and $C2$ denote the number of common EEG channels across patients in each dataset, while the dimensions $F \times T$ correspond to the frequency-time resolution of a 30-second time window.

2.3. Network architecture

As shown in Fig. 1, Module B is the "causal attention network with time frequency channel feature fusion". The data processed by Module A is used as the input for Module B. Module B is composed of time frequency channel feature fusion and CANet. The time frequency channel feature fusion part uses dilated causal convolution to perform local feature extraction and fusion in the time–frequency domain, followed by a convolutional layer that integrates features from all channels. In the CANet part, causal attention is used to extract global causal features, and then processed through two fully connected layers to finally output a series of labels for predicting the interictal or preictal period for Module B.

2.3.1. Time frequency channel feature fusion

Traditional manual feature extraction requires a lot of time and professional knowledge. Moreover, it may not be able to discover some hidden features in the data (Jia et al., 2016). Therefore, this paper uses automatic feature extraction. Recently, the temporal convolutional network (TCN) with a core component of dilated causal convolutional networks (Ji et al., 2024) has been employed to predict epilepsy, with numerous studies demonstrating its effectiveness in forecasting seizures. As shown in Fig. 2.

The dilated causal convolution incorporates dilation rates to increase spacing between kernel elements, thereby substantially expanding the receptive field without increasing parameter count or computational cost, enabling efficient processing. Through hierarchical stacking of layers with progressively varied dilation rates, this approach

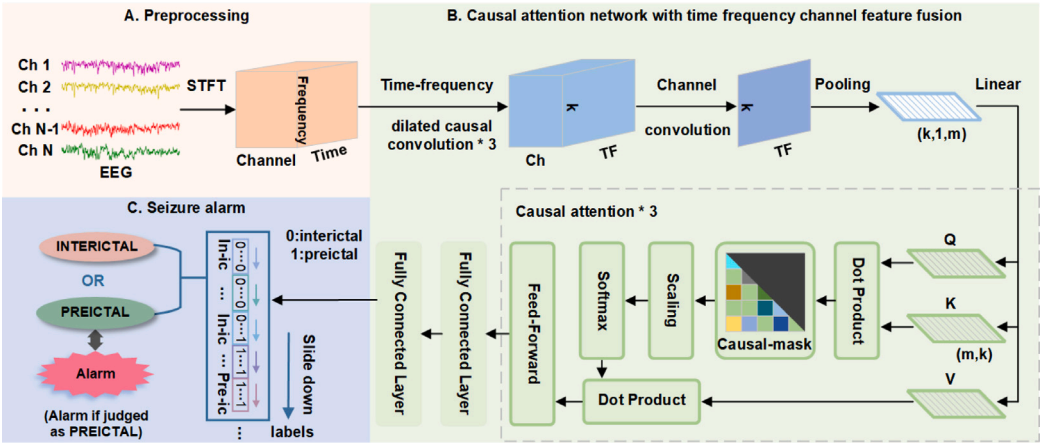

Fig. 1. Overview of the CANet. Where Ch denotes Channel, TF denotes Time–frequency, In-ic and Pre-ic respectively represent interictal and preictal period, while INTERICTAL and PREICTAL respectively denote interictal pre-ictal data segmentation.

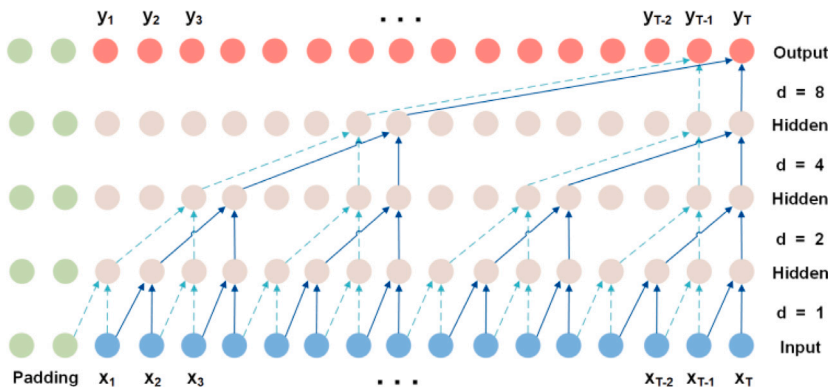

Fig. 2. Structural diagram of the dilated causal convolutional networks.

supports multi-scale feature extraction with exponentially growing receptive fields, while preserving temporal causality by relying solely on past inputs. This capability makes it particularly effective in capturing long-range pathological correlations in time series applications such as epilepsy prediction. The dilated causal convolutional network F can be formulated as:

$$\mathbf{F}(s) = (X *_d f)(s) = \sum_{i=0}^{k-1} f(i) \cdot X_{s-d\times i} \tag{2}$$

Here, s is the input sequence index, $X \in \mathbb{R}^N$ is the input sequence, $*_d$ is the dilated causal convolution operation and d is the dilation rate, $f : 0, \ldots, k-1 \rightarrow \mathbb{R}$ represents the filter and k is the filter size. This operation exhibits causal relevance, as the output at the current point s depends solely on the input data from the current point s and earlier points (that is, $X_{s-d\times i}$, $i \geq 0$), thereby ensuring no leakage of information about future points. This extension mechanism enables the model to capture dependencies along this dimension. In our architecture, the input to the dilated causal convolutional network in the Freiburg dataset is the STFT output shaped as $C1 \times TF$, where $C1$ denotes the number of electrode channels and TF represents the time–frequency data integration (i.e., 114 time frames × 61 frequency bins). The output data format is $H_K \times C1 \times TF$, where H_K denotes that the hyperparameter k in the dilated causal convolutional network is 16, and $C1 \times TF$ preserves the temporal and spectral resolution of the input. In this procedure, the CHB-MIT dataset differs from the Freiburg dataset in that it employs 18 electrode channels. This configuration is crucial to maintain the consistency of the dimensions in the data, enabling the extraction of multiscale features from epileptic EEG signals while preserving spatiotemporal information.

We use dilated causal convolutional networks for automatic feature extraction and fusion at the time and frequency levels, and standard convolution for feature fusion at the channel level. Inspired by Bi et al. (2024) and Xu et al. (2018), we separated the two-dimensional convolution into two one-dimensional convolution layers operating along distinct directions. We refer to Bi et al. (2024) for the methodology on flattening 2D spectrograms into 1D time–frequency data. Time–frequency domain feature extraction is first performed automatically using dilated causal convolution with k convolution kernels of size $(1, 128)$, where the step size is 2^{n-1} (n is the number of layers of the dilated causal convolution). We have shown experimentally that three time–frequency causal convolutional network layers have the best performance, so we finally use a three-layer time–frequency causal convolutional network. The second step involves applying k convolution kernels of size $(ch, 1)$ and stride $(1, 1)$ to fuse the channel features. Here, ch represents the number of electrode channels in the EEG data. Subsequently, we use batch normalization to enhance training stability and reduce the risk of overfitting. Following the method outlined in Shah et al. (2016), we selected the activation function of the Exponential Linear Unit (ELU) to enhance the nonlinear expression ability of the model. In the subsequent layer implements we utilized a kernel size of $(1, 128)$ and a stride of $(1, 32)$ for the average pooling operation along the time–frequency dimension. This pooling mechanism serves to smooth the time–frequency features, thus mitigating overfitting and lowering computational demands. Subsequently, the time–frequency and channel features are restructured: the electrode channel dimension is flattened and transposed with respect to the time–frequency dimension. As described in Table 3, the hyperparameter k is configured to 16. The time–frequency and channel features corresponding to each time point are fed into the CANet as input.

Table 3
Details of causal attention with time frequency channel feature fusion.

Layer	Layer number	In	Out	Kernel/Value	Stride
TF dilated causal Conv	3	$1(k)$	k	(1128)	$(1, 2^{n-1})$
Batch normalization		k	k	–	–
Elu activation				–	–
Average pooling				(1128)	$(1, 32)$
Dropout				0.5	–
Channel conv	1	k	k	$(ch, 1)$	$(1, 1)$
Batch normalization				–	–
Elu activation				–	–
Average pooling				(1128)	$(1, 32)$
Dropout				0.5	–
Rearrange	1	$(k, 1, m)$	(m, k)	–	–
Layer normalization	3	(m, k)	(m, k)	Embedding Size = 8	–
Causal attention				Embedding Size = 8; attention heads = 1; Dropout = 0.6; block size = 45	
Layer normalization				Embedding Size = 8	
Feedforward				Forward expansion = 1; Dropout = 0.5; Embedding Size = 8	
Fully connected	2	(m, k)	2	–	-

2.3.2. Causal attention network

The coherence of epileptic neural activity makes both time–frequency and channel features critical for epilepsy classification and prediction. This paper employs causal attention to capture global causal correlations in the fused time–frequency and channel features, while also addressing the limitations of constrained receptive fields in dilated causal convolutional networks. The tokens arranged in the time–frequency channel feature fusion section are linearly transformed to produce equal-sized triples called queries (Q), keys (K), and values (V). we use the dot product between Q and the transposed vectors of K to assess the relationship among distinct tokens. To uphold the causality principle within the model, we introduce a masking matrix M that conceals subsequent information frames beyond the current point. This masking is applied subsequent to the scaled dot product operation with the factor d. The attention mechanism utilizes the $Softmax$ function to compute attention weights (Fan et al., 2022). To prevent the model from accessing future information, a masking matrix is applied, where the upper triangular part (corresponding to future positions) is filled with $-\infty$, and the lower triangular part (representing the current and past positions) is set to 1. This strategic masking ensures that the model only attends to available, non-future data, thus preserving causality. A scaling factor is meticulously crafted to circumvent the issue of vanishing gradients, thereby ensuring a stable training regimen. This output is then funneled through a $Softmax$ function, yielding a weighting matrix that represents the causal attention scores, effectively capturing the significance of each token interaction. Subsequently, the causal attention scores are applied to the values (V) through dot product operations, thus effectively regulating the contribution of each feature dimension (Fan et al., 2022). This procedure can be encapsulated mathematically as:

$$CA(Q, K, V) = \text{Softmax}\left(M \cdot \frac{QK^T}{\sqrt{k}}\right)V \tag{3}$$

$$M(p, q) = \begin{cases} 1, & p \le m - q, \\ -\infty, & \text{otherwise.} \end{cases} \tag{4}$$

Here, k is a token's length, CA is causal attention, Q, K, and V denote the query, key, and value matrices, respectively. $\text{Softmax}\left(M \cdot \frac{QK^T}{\sqrt{k}}\right)$ is applied to row vectors, converting the log-likelihood values from the attention layer into a probability distribution. M is the causal masking matrix. p and q represent the horizontal and vertical coordinates of the causal masking matrix. The input to the causal attention module is a tensor of shape $m \times k$. This operation produces an output of the same dimensions $m \times k$, integrating global information from past and present data while preserving the sequence length and feature size.

Furthermore, we employed two fully connected feedforward layers to conduct deep global semantic modeling. The dimensions of the input and output data are maintained throughout the process. In the causal attention network, we defined the number of causal attention layers as three. The detailed parameters of the causal attention network are presented in Table 3.

Finally, two fully connected layers are employed, followed by a softmax activation function that transforms the output into a probability distribution, resulting in a one-dimensional predicted classification vector. The cross-entropy is used as the loss function to evaluate the error between the model's predicted output and the true labels (Wang et al., 2019), as shown below:

$$\mathcal{L} = -\frac{1}{N_b}\sum_{i=1}^{N_b}\sum_{c=1}^{M} y\log(\hat{y}) \tag{5}$$

Here, M is the classification number of epilepsy, y is the truth label, $\hat{y}$ is the predicted label, and N_b is the number of trials for a batch.

2.4. Seizure alarm

Frequent alarms may impose significant psychological stress on patients, interfere with their normal behaviors, and potentially result in unnecessary medical interventions. Thus, we introduce a double-layer dynamic window prediction algorithm to minimize false alert. The method is presented in Module C of Fig. 1, the corresponding mathematical formulation are outlined below. The sliding window mechanism progresses sequentially from interictal to preictal states. The first-level window encompasses 4 labels (as illustrated by the five colored small boxes in Fig. 1), here, label 0 is the interictal state, 1 is the preictal state; 4 labels represent 2 min of data. If the count of label 1 within this window does not exceed 2 ($n1 \le P1$), the window is classified as In-ic; otherwise, it is labeled as Pre-ic. The second-level window spans 20 labels (represented by the large blue box), which corresponds to five consecutive first-level windows or 10 min of data. When the number of Pre-ic classifications within this window is no more than 3 ($n2 \le P2$), the window is categorized as INTERICTAL; otherwise, it is designated as PREICTAL. This second-level window then advances non-overlappingly along the interictal-to-preictal direction, and an alarm is activated upon its classification as PREICTAL.

$$\text{State1}[t] = \begin{cases} \text{In-ic}, & n1 \le T1, \\ \text{Pre-ic}, & n1 > T1. \end{cases} \tag{6}$$

$$\text{State2}[T] = \begin{cases} \text{INTERICTAL}, & n2 \le T2, \\ \text{PREICTAL}, & n2 > T2. \end{cases} \tag{7}$$

Here, 0 is predicted interictal label; 1 is predicted preictal label; $n1$ is the count of label 1 in the first-level window; $T1$ is the threshold of 2; $n2$ is the number of Pre-ic labeled within the second-level window; $T2$ is the threshold of 3.

2.5. Experimental strategy

This study employed the leave-one-out cross-validation (LOOCV) strategy (Gao et al., 2022), and "one" refers to a single seizure event. Specifically, for a patient with N seizure episodes (with the pre-ictal period corresponding to 1 or 2 h), we randomly selected N interictal segments of the same quantity as the pre-ictal periods to ensure class balance during model training (Wang et al., 2021). The interictal and pre-ictal periods were randomly combined in pairs (left: interictal, right: pre-ictal) to form N paired segments. In each epoch, one segment was selected as the test set, and the remaining N-1 segments were used as the training set until each segment was considered as a test data once. Therefore, our model was experimented N times for each patient, outputting a series of labels indicating the interictal or pre-ictal period. We calculate the average accuracy, sensitivity and false positive rate as the final results. The entire experiment was repeated six times, generating six sets of predicted labels, each of which was input into module C (seizure alert) in Fig. 1.

For the two datasets employed in our study, the parameters of the model were set to batch size of 32, learning rate of 0.0005, embedding dimension of 8, epochs of 200, the number of layers of dilated causal convolution was set to 3, and the causal attention mechanism was configured with 3 layers. The model employed in this study was developed using Python 3.10 and executed on an NVIDIA GTX 4090 GPU. The model contains approximately 5M parameters, with a computational complexity of about 11 GFLOPs per sample. For instance, training time in a 1-hour preictal experiment on the Freiburg dataset was approximately 2 min.

3. Results

In this section, we first introduce the performance evaluation criteria based on segments and events. Then, we present the results on the two datasets we chose, including detailed performance of our seizure prediction method and comparisons showing how key components affect overall system effectiveness.

3.1. Evaluation metrics

In seizure prediction, approaches are typically evaluated at the event-based level. Simultaneously, we conducted experiments at the segment-based level to more thoroughly assess method performance and the differences between event-based and segment-based evaluations.

Before running the algorithm, the SOP and the Seizure Prediction Horizon (SPH) should be defined (Maiwald et al., 2004). At the event-based level, SPH is defined as the time interval between the system's alert trigger and the actual onset of an epileptic seizure. SOP typically represents equivalent to the preictal interval (Hu et al., 2023; Rao et al., 2024; Lopes et al., 2023). A longer interictal and preictal period is more conducive to our experimental research. Therefore, we chose a longer interictal and preictal of 1 and 2 h respectively as the research object. Thereby, the SOP is set to 1 and 2 h, respectively. We used a longer interictal and preictal period in our experiment, since extended data segments allow for more thorough learning, a relatively long SPH was also employed. In this study, the SPH was defined as 10 min. The SEN and FAR was employed to evaluate our prediction model. as explained in the following sections.

SEN is the percentage of correctly predicted warnings among all seizure episodes. It is defined as:

$$SEN = \frac{cor}{N} \times 100\% \tag{8}$$

Here, cor is the total count of accurate alerts; N is the total number of episode seizures.

FAR indicates total false alarms produced by the epilepsy alert system per hour. It is defined as:

$$FAR = \frac{fal}{T} \tag{9}$$

Here, fal is the total count of false alarms; T denotes the overall duration hours of the EEG signal.

At the segment-based level, the accuracy(Acc), sensitivity(Sen), specificity(Spe), and false prediction rate(Fpr) are employed as evaluation metrics. The formulas of these four metrics are given as follows:

$$Acc = \frac{TP + TN}{TP + TN + FN + FP} \tag{10}$$

$$Sen = \frac{TP}{TP + FN} \tag{11}$$

$$Spe = \frac{TN}{TN + FP} \tag{12}$$

$$Fpr = \frac{FP}{TN + FP} \tag{13}$$

Here, TP (True Positive): The number of windows correctly labeled as preictal; FP (False Positive): The number of windows wrongly labeled as preictal when they are actually interictal; TN (True Negative): The number of windows correctly labeled as interictal; FN (False Negative): The number of windows wrongly labeled as preictal when they are actually interictal.

3.2. Experimental results

3.2.1. The experiments on freiburg dataset

We evaluated the effectiveness of our proposed approach at both the segment-based and event-based levels with a 1-hour preictal interval durations, as presented in Table 4. On the segment-based level, our method achieved an Acc of 96.12%, a Sen of 93.87%, a Spe of 98.37%, and a Fpr of 0.0163/h. Among 20 patients, 17 achieved over 90.00% Acc with our method, and for half of the patients, the Acc exceeded 99.00%. For Patient 4, although the Sen was less than ideal, the Spe was very high, indicating that the method was less effective at predicting the preictal period but performed well at identifying the interictal periods. The main reason for this is that the patient's partial preictal is too close to the interictal state, resulting in a lower overall Sen in this patient. For Patient 20, our method exhibited mediocre performance compared to other patients, with a Sen of 80.33% and a Fpr of 0.1217/h. This suggests that, for this specific patient, our method exhibited a diminished ability to distinguish between preictal and interictal phases. On the event-based level, our method achieved a 100.00% SEN and a FAR of 0.0077/h. This level of sensitivity indicates that the method can effectively predict each epileptic seizure, enabling patients to take necessary measures before the seizure occurs, thereby alleviating or reducing the serious consequences brought about by epilepsy. Among these patients, 80% had a FAR of 0, indicating that our method produced no false alarms for them, which significantly reduced patient anxiety. The results at both the segment-based and event-based levels for the Freiburg dataset with a preictal interval durations of 1 h demonstrate that our method effectively identifies preictal and interictal periods and exhibits excellent performance in seizure prediction.

Moreover, we evaluated the effectiveness of the 2 h preictal interval durations, both at the segment-based and event-based levels, as presented in Table 5. At the segment-based level, our method achieved an Acc of 91.89%, Sen of 89.36%, Spe of 94.42%, and a Fpr of 0.0558/h. Among the 15 patients, 10 exhibited an Acc of over 95.00% with our method. Our method struggled to accurately identify the preictal and interictal periods for patients 4 and 19. Specifically, for patient 4, the Sen was 59.79%, indicating difficulty in determining the true preictal period for this patient. For patient 19, the Fpr was 0.4279/h, reflecting challenges in identifying the true interictal period. At the event-based level, our method achieved a SEN of 96.67% and a FAR

Table 4
Freiburg dataset: predictive performance with a preictal interval duration of 1 h.

Patient ID	No. of seizures	Segment-based level				Event-based level	
		Acc (%)	Sen (%)	Spe (%)	Fpr (/h)	SEN (%)	FAR (/h)
Burg01	3	92.64	86.94	98.33	0.0167	100.00	0.0000
Burg03	4	99.17	98.61	99.72	0.0028	100.00	0.0000
Burg04	4	81.67	63.43	99.91	0.0009	100.00	0.0000
Burg05	2	98.33	99.68	96.99	0.0301	100.00	0.0278
Burg06	3	99.17	98.89	99.44	0.0056	100.00	0.0000
Burg07	3	96.81	94.44	99.17	0.0083	100.00	0.0000
Burg08	2	99.17	98.52	99.81	0.0019	100.00	0.0000
Burg09	5	99.83	99.75	99.92	0.0008	100.00	0.0000
Burg10	5	97.08	95.16	99.01	0.0099	100.00	0.0000
Burg11	4	99.58	99.38	99.79	0.0021	100.00	0.0000
Burg12	4	97.81	95.75	99.87	0.0013	100.00	0.0000
Burg13	2	88.33	82.50	94.17	0.0583	100.00	0.0417
Burg14	3	99.86	99.98	99.74	0.0026	100.00	0.0000
Burg15	3	97.08	98.61	95.56	0.0444	100.00	0.0185
Burg16	5	99.75	99.91	99.59	0.0041	100.00	0.0000
Burg17	5	99.92	99.98	99.85	0.0015	100.00	0.0000
Burg18	5	99.75	99.67	99.83	0.0017	100.00	0.0000
Burg19	4	99.27	99.17	99.38	0.0062	100.00	0.0000
Burg20	5	84.08	80.33	87.83	0.1217	100.00	0.0667
Burg21	4	93.13	86.74	99.51	0.0049	100.00	0.0000
Average	**75**	**96.12**	**93.87**	**98.37**	**0.0163**	**100.00**	**0.0077**

Table 5
Freiburg dataset: predictive performance with a preictal interval duration of 2 h.

Patient ID	No. of seizures	Segment-based level				Event-based level	
		Acc (%)	Sen (%)	Spe (%)	Fpr (/h)	SEN (%)	FAR (/h)
Burg01	2	88.75	78.33	99.17	0.0083	100.00	0.0000
Burg04	2	72.71	59.79	85.63	0.1437	50.00	0.1111
Burg05	2	97.81	97.71	97.92	0.0208	100.00	0.0000
Burg06	2	99.06	99.79	98.33	0.0167	100.00	0.0000
Burg09	2	98.02	99.92	96.13	0.0388	100.00	0.0278
Burg10	2	91.56	84.65	98.47	0.0153	100.00	0.0000
Burg11	2	95.94	94.13	97.75	0.0225	100.00	0.0139
Burg12	2	96.35	94.10	98.61	0.0139	100.00	0.0000
Burg15	3	99.65	99.70	99.61	0.0039	100.00	0.0000
Burg16	2	98.54	98.54	98.54	0.0146	100.00	0.0000
Burg17	4	96.67	94.76	98.58	0.0142	100.00	0.0069
Burg18	4	96.35	93.96	98.75	0.0125	100.00	0.0000
Burg19	2	61.56	65.92	57.21	0.4279	100.00	0.5139
Burg20	4	87.66	82.60	92.71	0.0729	100.00	0.0347
Burg21	2	97.71	96.49	98.93	0.0108	100.00	0.0000
Average	**37**	**91.89**	**89.36**	**94.42**	**0.0558**	**96.67**	**0.0472**

of 0.0472/h. Out of 37 seizures in this dataset, 36 were predicted by our method. Notably, 60% of the patients had a FAR of 0, indicating no false alarms and thereby significantly reducing patient anxiety. The SEN for patient 4 is 50.00%, which means that only one out of the patient's two seizures can be correctly predicted. For patient 19, the FAR was 0.5139/h, indicating a high false alarm rate for this patient. The results indicate that performance at the segment-based level is generally aligned with that at the event-based level, though the latter consistently demonstrates superior performance. Overall, our method shows high SEN and low FAR at the event-based level. Comparing the 1-hour and 2-hour preictal intervals durations that the 1-hour interval durations better overall results. Nevertheless, this comparison also demonstrates that our method performs well over longer prediction horizons.

3.2.2. The experiments on CHB-MIT dataset

We effectiveness the performance of our approach at a 1-hour preictal interval, both at the segment-based and event-based levels, as presented in Table 6. At the segment-based level, our method achieved an Acc of 95.09%, a Sen of 93.95%, a Spe of 96.23%, and a Fpr of 0.0377/h. For 12 out of 17 patients, our method achieved an Acc greater than 95.00%. Only patient 2 had a Sen below 80.00%, with a SEN of just 50.00% at the event-based level, indicating that our method does not effectively identify the true interictal periods for this patient. At the event-based level, our method achieved a SEN of 97.06% and a FAR of 0.0251/h. Our method successfully predicted 64 out of 65 seizures in this dataset, demonstrating its strong predictive capability. Furthermore, for 11 out of 17 patients, the FAR was 0, indicating that our method does not produce false alarms for these individuals. Overall, the experimental results show that our method also performs well in predicting seizures at a 1-hour preictal interval durations in the CHB-MIT dataset.

Furthermore, in the CHB-MIT dataset, we evaluated the performance of our method for seizure prediction at a 2-hour preictal interval, both at the segment-based and event-based levels, as shown in Table 7. At the segment-based level, our method achieved an Acc of 86.10%, a Sen of 80.63%, a Spe of 91.58%, and a Fpr of 0.0842/h. For 5 out of 13 patients, our method reached an Acc of over 95.00%. However, for patients 2 and 13, the prediction results were not satisfactory. Specifically, the Sen for patients 2 and 13 were 57.08% and 59.54%, respectively, and at the event-based level, both patients had a SEN of 50.00%, indicating that our method does not effectively identify the true preictal periods for these patients. At the event-based level, our method achieved a SEN of 92.31% and a FAR of 0.0666/h. For 11 out of 13 patients, our method achieved a SEN of 100.00%, meaning it accurately predicted seizures for these individuals. Overall, while the

Table 6
CHB-MIT dataset: predictive performance with a preictal interval duration of 1 h.

Patient ID	No. of seizures	Segment-based level				Event-based level	
		Acc (%)	Sen (%)	Spe (%)	Fpr (/h)	SEN (%)	FAR (/h)
Chb01	5	99.75	99.67	99.83	0.0017	100.00	0.0000
Chb02	2	80.16	60.45	99.88	0.0012	50.00	0.0000
Chb03	4	92.40	86.88	97.92	0.0208	100.00	0.0000
Chb04	3	97.77	98.60	96.94	0.0306	100.00	0.0185
Chb05	5	99.00	99.33	98.67	0.0133	100.00	0.0000
Chb06	9	83.10	82.96	83.24	0.1676	100.00	0.1481
Chb07	3	88.03	97.83	78.22	0.2178	100.00	0.1667
Chb09	4	99.72	99.94	99.51	0.0049	100.00	0.0000
Chb10	6	93.54	90.42	96.67	0.0333	100.00	0.0417
Chb13	3	98.19	97.20	99.19	0.0081	100.00	0.0000
Chb16	3	98.58	99.78	97.39	0.0261	100.00	0.0000
Chb18	3	97.14	99.31	94.97	0.0503	100.00	0.0370
Chb19	3	96.11	92.84	99.39	0.0061	100.00	0.0000
Chb20	3	99.86	99.91	99.81	0.0019	100.00	0.0000
Chb21	3	96.81	98.05	95.56	0.0444	100.00	0.0139
Chb22	3	97.50	95.42	99.59	0.0042	100.00	0.0000
Chb23	3	98.88	98.60	99.17	0.0083	100.00	0.0000
Average	**65**	**95.09**	**93.95**	**96.23**	**0.0377**	**97.06**	**0.0251**

Table 7
CHB-MIT dataset: predictive performance with a preictal interval duration of 2 h.

Patient ID	No. of seizures	Segment-based level				Event-based level	
		Acc (%)	Sen (%)	Spe (%)	Fpr (/h)	SEN (%)	FAR (/h)
Chb01	5	99.75	99.60	99.90	0.0010	100.00	0.0000
Chb01	5	99.75	99.60	99.90	0.0010	100.00	0.0000
Chb02	2	66.15	57.08	75.21	0.2479	50.00	0.2917
Chb04	3	97.08	95.57	98.61	0.0139	100.00	0.0000
Chb05	4	97.71	99.01	96.40	0.0360	100.00	0.0208
Chb06	6	80.83	77.29	84.38	0.1563	100.00	0.1111
Chb07	3	80.76	71.84	89.69	0.1031	100.00	0.0833
Chb09	3	99.28	98.92	99.64	0.0036	100.00	0.0000
Chb10	6	72.19	62.71	81.67	0.1833	100.00	0.1667
Chb13	3	73.54	59.54	87.54	0.1246	50.00	0.1042
Chb16	2	89.38	83.04	95.71	0.0429	100.00	0.0000
Chb18	2	85.63	77.13	94.13	0.0588	100.00	0.0208
Chb20	2	79.43	68.37	90.48	0.0952	100.00	0.0667
Chb22	3	97.60	98.05	97.15	0.0285	100.00	0.0000
Average	**44**	**86.10**	**80.63**	**91.58**	**0.0842**	**92.31**	**0.0666**

performance of our method at a 2-hour preictal interval was not as good as at a 1-hour interval, the results are still considered satisfactory.

From the above four experiments, it is evident that the results at a 1-hour preictal interval are superior to those at a 2-hour preictal interval. However, for patients, the results at a 2-hour preictal interval are more valuable. A longer prediction horizon provides the potential to prevent seizures through the use of antiepileptic drugs, which can help patients avoid the severe consequences of seizures. The overall results from the Freiburg dataset are better than those from the CHB-MIT dataset, indicating that our method performs better with intracranial EEG data.

3.2.3. The experiments on APT prediction

Following the methods (Li et al., 2021; Gao et al., 2023; Guo et al., 2023), We use APT (Average Prediction Time, the mean interval between our system alert and actual epilepsy onset) to evaluate whether our method predicts epilepsy effectively and in a timely manner. An effective seizure prediction method should have a sufficiently long APT to help patients get enough time to take measures. Specifically, antiepileptic drugs can effectively prevent epilepsy. Timely and accurate epilepsy warnings can help those patients who need to take medication at regular intervals every day, potentially reducing the reliance on routine medication. This requires that our APT time exceeds the onset time of conventional antiepileptic drugs. Our goal is to explore longer APT, and with a larger SOP setting, the APT is relatively larger. Therefore, we only analyze the APT in experiments with a 2-hour preictal interval durations (i.e., SOP equals 2 h). The APT results of our approach on the Freiburg and CHB-MIT datasets are shown in Fig. 3.

As illustrated in Fig. 3, on the Freiburg dataset, our method achieved an average APT of 97.59 min. Among the 15 patients, patients 4 and 19 had lower APT but still exceeded 70.00 min. Thirteen out of the 15 patients had APT greater than 90.00 min. On the CHB-MIT dataset, our method achieved an average APT of 94.85 min. Patients 2, 10 and 13 had lower APT but still exceeded 60.00 min. Nine out of the 13 patients had APT greater than 90.00 min. Most antiepileptic drugs require more than 30.00 min to become effective, with some even taking over an hour (Johannessen Landmark et al., 2020; Meirinho et al., 2021; Patsalos et al., 2008). The experimental results indicate that our method provides an APT greater than 60.00 min for all patients in both the Freiburg dataset and CHB-MIT dataset, allowing for the timely administration of antiepileptic drugs with a faster onset time to prevent seizures. Moreover, with over 60% of patients in both datasets having an APT exceeding 90.00 min, our method for seizure prediction allows these patients to administer most antiepileptic drugs in a timely manner, thereby helping to avert seizures.

3.2.4. Ablation experiments

To better validate the impact of different modules on the overall model, we conducted a series of ablation experiments. The STFT transforms time-domain EEG signal data into time–frequency domain data. To validate the effectiveness of the STFT module within the overall model, we conducted ablation experiments with and without STFT on two datasets and at the preictal periods of 1 h and 2 h, as shown in

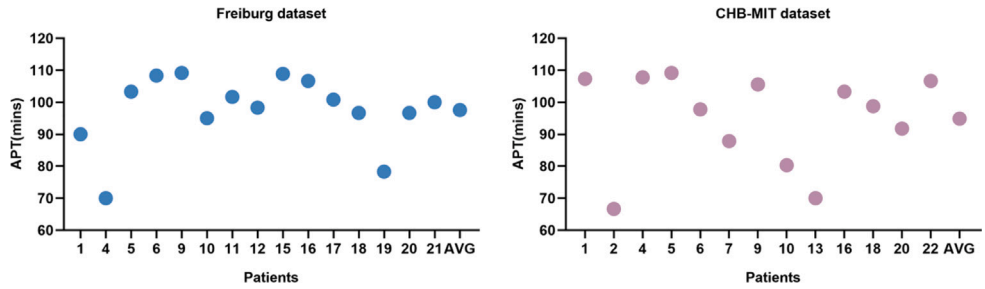

Fig. 3. Results of the APT in the Freiburg dataset and the CHB-MIT dataset.

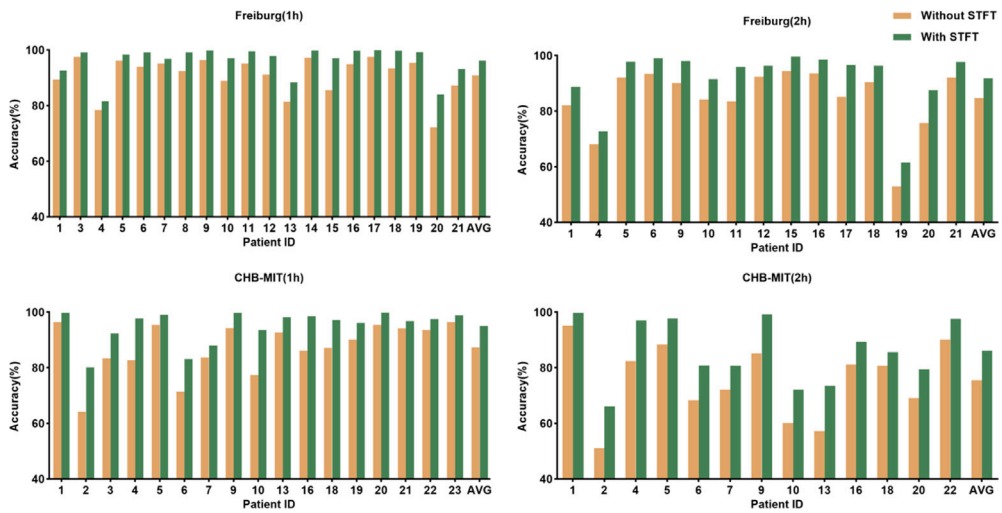

Fig. 4. Comparison of accuracy results with and without STFT in the preictal interval durations of 1 and 2 h on two datasets.

Fig. 4. In the Freiburg and CHB-MIT datasets, the average accuracy was higher for models with the STFT module compared to those without it. Furthermore, for each patient, the average accuracy was higher for models incorporating the STFT module than for those without it. Whether in the 1-hour or 2-hour preictal period experiments, the CHB-MIT dataset showed a greater gap between the Freiburg dataset from the with and without STFT modules. Specifically, in the 2 h of preictal period data from the CHB-MIT dataset, the average accuracy without the STFT module was 75.50%, which was 10.60% lower than the average accuracy with the STFT module. Therefore, multiple experiments have demonstrated that incorporating an STFT module can effectively enhance the overall model performance.

To further validate the effectiveness of the STFT module, we conducted paired t-tests to analyze the statistical significance of the accuracy differences between the "without STFT" and "with STFT" conditions. The results show that under all experimental conditions (including burg_1h, burg_2h, chb_1h, and chb_2h), the accuracy differences reached a statistically highly significant level ($p < 0.001$), as shown in Table 8. All effect sizes (Cohen_d) were greater than 1.64, indicating that the performance improvement brought by the STFT module is of substantial practical significance. These statistical results further confirm that incorporating the STFT module into the model can significantly and substantially enhance the performance.

We explored the performance of the dilated causal convolution networks (DCCN) with and without its application across different datasets and for preictal interval durations of 1 and 2 h. The event-level results of our method show a strong alignment with the segment-level outcomes, as demonstrated by the various metrics and performance indicators. Specifically, better results at the event-based level are associated with better outcomes at the segment-based level. Since the accuracy metric at the segment level is inferred from other metrics, it effectively evaluates the performance of our method. To clearly illustrate the performance differences with and without the component, we used only the accuracy metric for comparison. We compared the accuracy results with and without DCCN for preictal interval durations of 1 and 2 h across two datasets, as illustrated in Fig. 5.

Table 8
Details of the statistical significance with and without STFT.

Experiment	*t-Statistic*	*p-Value*	Cohen_*d*
burg_1h	8.23	7.43E−08	1.80
burg_2h	10.74	1.93E−08	2.69
chb_1h	6.971	2.25E−06	1.64
chb_2h	10.77	7.55E−08	2.88

The experimental results in Fig. 5 demonstrate that for both datasets used and for preictal periods of 1 h and 2 h, the average accuracy achieved with DCCN surpasses that obtained without DCCN. In the CHB-MIT dataset the performance gap between the with and without of DCCN is greater than in the Freiburg dataset, indicating a more substantial impact of DCCN on data collected from the scalp. In the Freiburg dataset, for patients with a 1-hour preictal interval durations, the Acc without DCCN is below 70.00%, whereas with DCCN, it exceeds 80.00%, highlighting the significant benefit of DCCN for these patients. For patients with 2-hour preictal interval durations, DCCN also contributes significantly to accuracy improvement. Similarly, in the CHB-MIT dataset, some patients with 1 and 2-hour preictal interval durations also experience a notable impact from DCCN. Overall, across all datasets and patients, accuracy with DCCN is consistently higher than without, demonstrating that DCCN improves the comprehensive effectiveness.

Paired t-tests was performed to validate the effectiveness of the DCCN module. The result as shown in Table 9 confirmed that integrating the DCCN module significantly improved model accuracy across

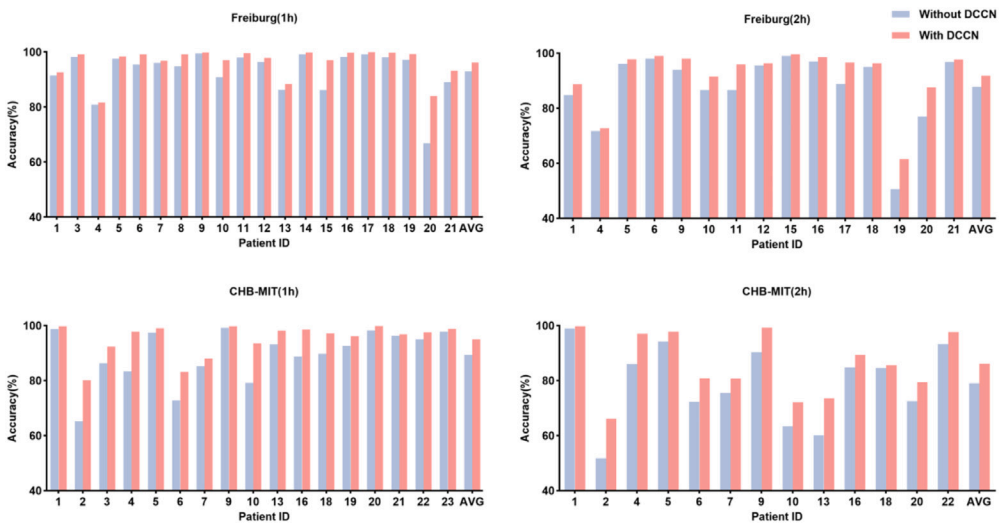

Fig. 5. Comparison of accuracy results with and without DCCN in the preictal interval durations of 1 and 2 h on two datasets.

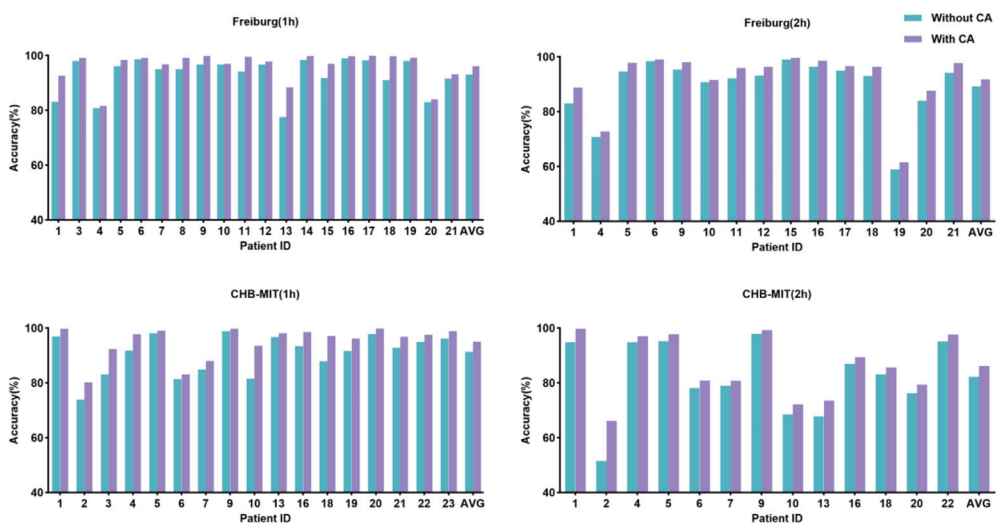

Fig. 6. Comparison of accuracy results with and without CA in the preictal interval durations of 1 and 2 h on two datasets.

Table 9
Details of the statistical significance with and without DCCN.

Experiment	*t-Statistic*	*p-Value*	Cohen_*d*
burg_1h	3.61	1.75E−03	0.79
burg_2h	6.6	8.48E−06	1.65
chb_1h	4.81	1.62E−04	1.13
chb_2h	6.34	2.56E−05	1.69

all experimental conditions (all $p < 0.001$). The improvements were substantiated by large to very large effect sizes (Cohen_d: 0.79–1.69), indicating both statistical significance and practical relevance. These results demonstrate that the DCCN module consistently enhances the performance.

Causal attention (CA) is another crucial component of our method. The objective of this study is to assess how the presence or absence of CA affects performance across various datasets for preictal interval durations of 1 and 2 h. We compared accuracy results with and without CA on both the two datasets for preictal interval durations of 1 and 2 h, as shown in Fig. 6.

The experimental results in Fig. 6 demonstrate that for both datasets used and for preictal periods of 1 h and 2 h, the average accuracy achieved with CA is higher than that without CA. The performance gap between with and without CA is relatively consistent across datasets, indicating a similar impact of CA on intracranial and scalp EEG data. For patients 1 and 13 in the Freiburg dataset with a preictal interval durations of 1 h, CA contributes significantly to accuracy improvement. Similarly, in the CHB-MIT dataset, some patients show substantial benefits from CA for preictal interval durations of 1 and 2 h. Overall, across all patients and datasets, accuracy with CA consistently exceeds accuracy without CA, demonstrating that CA improves the comprehensive effectiveness.

Statistical analysis with and without CA confirmed that integrating the CA module significantly improved model accuracy across all experimental conditions (all $p < 0.001$), and Cohens_d in 1.01–1.98, as shown in Table 10. These results demonstrate that the CA module consistently enhances the performance.

Finally, we explored the impact of varying the number of dilated causal convolutions (DCCs) on the overall method and investigated the performance differences among different numbers of DCCs for preictal interval durations of 1 and 2 h across different datasets. We compared accuracy results for different numbers of DCCs on the two datasets for preictal interval durations of 1 and 2 h, as shown in Fig. 7. To more clearly illustrate the trend in results across different numbers of DCCs used, the accuracy results in this experiment represent the average accuracy across all patients.

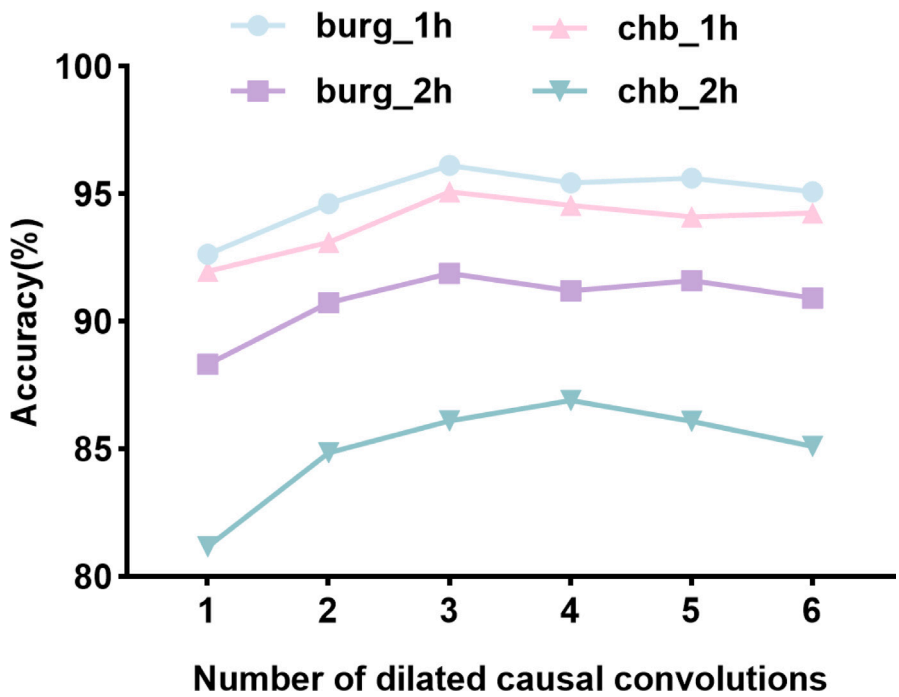

Fig. 7. Comparison of accuracy results for different numbers of dilated causal convolutions in the preictal intervals of 1 and 2 h on two datasets.

Table 10
Details of the statistical significance with and without CA.

Experiment	*t-Statistic*	*p-Value*	Cohen_*d*
burg_1h	4.61	1.70E−04	1.01
burg_2h	7.93	9.64E−07	1.98
chb_1h	5.89	1.80E−05	1.39
chb_2h	4.35	7.93E−04	1.16

Fig. 7 shows that for 1 to 3 DCCs, whether in the Freiburg or CHB-MIT datasets and for both preictal interval durations of 1 and 2 h, accuracy generally increases with the number of DCCs. However, for 3 to 6 DCCs, there is some fluctuation in the Freiburg dataset, but the overall trend indicates that accuracy decreases with more DCCs. A preictal interval durations of 2 h in the CHB-MIT dataset, accuracy is highest with 4 DCCs. Considering the results across different datasets and preictal interval durations, our final method uses 3 DCCs, meaning the DCCN is composed of three layers of DCCs.

4. Discussion

Epilepsy patients suffer both physically and mentally, and bear a heavy economic burden. Better epilepsy prediction can help patients reshape their lives. The method we proposed use dilated causal convolution and Causal attention fully capture the EEG signals, and a double-layer dynamic window can better predict seizure. This section discusses our proposed model, which extracts local features through the convolutional networks and global features through the causal attention networks. This section evaluates existing methods from segment-based and event-based views, highlighting the strengths of our approach. It also outlines current limitations and future research directions.

Primarily, to better explain the specific functions of multiple modules in the model, we will further discuss the convolutional network module and the causal attention network module. The convolutional network module consists of inflated convolutional networks and convolutional neural networks. Its core objective is to fuse local features across the time domain, frequency domain, and channel domain. Due to the inherent characteristics of convolutional networks, they primarily extract information from the current data and its surrounding context to continuously learn. Compared to attention methods, convolutional networks learn more localized data. However, their advantage is that it can focus more intently on the features of surrounding data, extracting more effective features from the neighborhood. To clearly demonstrate the local features extracted by the convolutional network, we present feature visualizations before and after the convolutional network module, as shown in Fig. 8A and 8B. The selected displayed data is the first test data segment from patient 1 in the Freiburg dataset. As shown in the figure, the convolutional network extracts a large number of data columns into dozens of columns, significantly enhancing feature discrimination compared to the input module. However, this module is limited to local feature extraction.

Causal attention networks can attend to all features, enabling them to effectively extract global features. The causal attention network module performs further global feature extraction on the pre-extracted local features, enabling the model to better capture effective features. Fig. 8B and 8C illustrate the feature changes before and after the causal attention network module. As shown in the figure, compared to the features before inputting the causal attention network module, the features after the causal attention network module are more significant. Moreover, the attention mechanism can effectively reference these features to better enhance the ability of epilepsy prediction.

Regarding the comparison between our approach and other methods, we discuss it at both the segmentation-based and event-based levels. At the segment-based level, comparing our method with the several existing advanced methods using the two datasets. Table 11 shows the detailed comparison. With respect to the precise differentiation between the interictal and preictal phases, the Freiburg dataset generally outperform those from the CHB-MIT dataset, indicating that intracranial EEG data can more effectively compared to scalp EEG data. In the Freiburg dataset, Zhang et al. (2015) used a gradient boosting method to achieve Acc, Sen, and Spe of 95.75%, 91.01%, and 95.77% respectively. This approach constructs a strong classifier from multiple weak classifiers, resulting in better classification. Geng et al. (2020) utilized BiLSTM and S-transform to achieve sensitivity and specificity of 98.09%, 98.69% respectively. This demonstrates that transforming EEG data into time–frequency features and applying deep learning

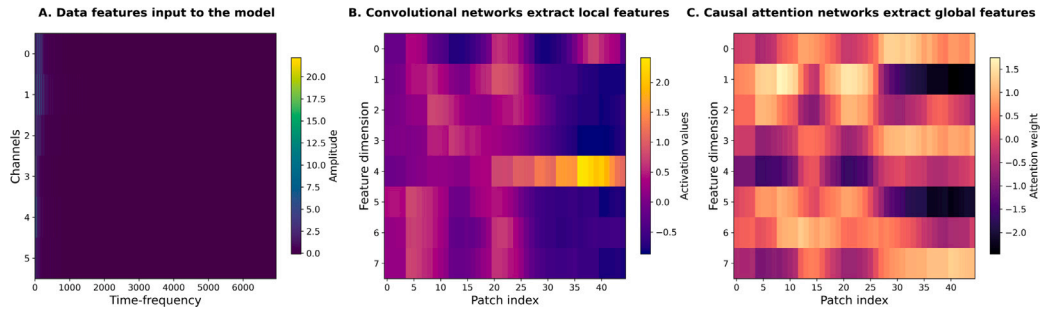

Fig. 8. Data features before and after modules.

Table 11
Segment-based level comparison with the existing advanced methods.

Authors	Year	Dataset	Method	Acc (%)	Sen (%)	Spe (%)	Fpr (/h)
Zhang et al. (2015)	2015		Gradient boosting	95.75	91.01	95.77	–
Zhou et al. (2018)	2018		CNN	96.70	96.70	96.80	–
Geng et al. (2020)	2020		BiLSTM with S-transform	–	**98.09**	**98.69**	–
Wang et al. (2022b)	2022	Freiburg	1D-CNN	95.13	86.23	96.00	–
Xu et al. (2023)	2023		CNN-BiGRU	**98.35**	93.68	–	0.3970
Wang et al. (2024)	2024		CNN-LSTM	93.50	88.10	94.00	–
Ours	2025		CANet	96.12	93.87	98.37	**0.0163**
Toraman (2020)	2020		CNN and SVM	91.05	92.32	89.76	–
Sun et al. (2021)	2021		CADCNN	–	**97.10**	95.60	**0.0290**
Yang et al. (2021)	2021		RDANet	92.07	89.33	93.02	–
Dissanayake et al. (2021)	2022	CHB-MIT	GDL	**95.38**	94.47	94.16	–
Assali et al. (2023)	2023		CNN	94.50	92.80	–	–
Ahmad et al. (2024)	2024		MCNNN-AT-Bi	91.39	91.30	90.06	0.1200
Ours	2025		CANet	95.09	93.95	**96.23**	0.0377

[1] Acc: Accuracy; Sen: Sensitivity; Spe: Specificity; Fpr: False prediction rate;
[2] Highlighted indicates the best-performing.

models improves differentiation between interictal and preictal stages, resulting in excellent epilepsy prediction performance. Zhou et al. (2018) achieved the Acc, Sen and Spe of 96.70%, 96.70%, 96.80% respectively, with a CNN method. Besides, some studies (Xu et al., 2023; Wang et al., 2024) have combined CNN with other methods, also attained favorable results. Similarly, in the CHB-MIT dataset, studies combined CNN with other methods achieved a maximum Sen and Spe of 97.10%, 95.60% respectively (Sun et al., 2021; Toraman, 2020; Ahmad et al., 2024). This indicates that appropriately combining multiple methods can leverage their respective advantages and enhance the performance of epilepsy prediction. Compared to those methods, our model effectively integrates various methods and makes full use of the features in both the time–frequency domain and the channel domain. Based on the evaluation using Acc, Sen, Spe, and Fpr, our model outperforms the existing advanced methods.

Subsequently, at the event-based level, we compared our approach with the existing advanced methods across the two datasets. Table 12 shows the detailed results. Epilepsy prediction research has primarily evolved along three methodological pathways: threshold analysis, machine learning, and deep learning. Hung et al. (2010) developed an autonomous prediction system based on wavelet transform and chaos theory, achieving a SEN of 86.96% and a FAR of 0.2500/h. Their key innovation was the introduction of nonlinear energy operators as novel features. Eftekhar et al. (2014) further enhanced performance by applying threshold-crossing techniques, yielding a SEN of 90.95% and reducing FAR to 0.0600/h, thereby demonstrating the robustness of threshold-based approaches. In studies (Sharif and Jafari, 2017; Yang et al., 2018), support vector machine (SVM) models achieved SEN between 91.80% and 96.60%, with FAR maintained within 0.0500–0.1100/h, underscoring their effectiveness in feature classification tasks. Recently, deep learning methods have gained prominence due to their superior feature extraction capabilities. As summarized in Table 12, LSTM- and CNN-based approaches report SEN ranging from 87.80% to 100.00%, with FAR varying between 0.0500 and 1.6000/h. Notably, Zhang et al. (2023) introduced the spatio-temporal causal neural network (STCNN), which employs dilated causal convolutions to model causal dependencies in EEG time series, delivering strong performance across multiple datasets. Despite these advances, limitations remain: STCNN does not incorporate advanced mechanisms such as attention modules, and its modeling of local and global causal relationships is insufficient. Future improvements could integrate graph neural networks and multi-modal data to enhance predictive accuracy.

However, it is meaningless that in our experiments where the preictal interval duration was set to 2 h, our method demonstrated strong performance with an APT exceeding that of commonly used antiepileptic drugs. This indicates that our approach holds promise for preventing seizures. Compared to previous studies, our method not only integrates features from different levels such as time–frequency and channels but also utilizes dilated causal convolutions to capture local causal features and causal attention to capture global causal features, thereby leveraging their respective advantages effectively. Based on the evaluation of SEN and low FAR, the experimental results show that the method we proposed has achieved better epilepsy prediction performance. As shown in Tables 11 and Tables 12, our approach outperforms most of the previous methods.

As to the Freiburg and CHB-MIT datasets, our method achieves high performance. Its APT surpasses the onset time of commonly used antiepileptic drugs, suggesting strong potential for effective seizure prevention. However, our proposed approach has not been evaluated the predictive performance on other datasets, and further validation is required to determine its generalization capability. In real-world applications, an increased number of electrodes may lead to greater patient discomfort. We have not explored strategies for achieving reliable epilepsy prediction using fewer electrodes while maintaining high performance. Current epilepsy seizure prediction systems do not meet the accuracy standards needed to generate reliable pre-seizure medication alerts. Additionally, there is still no clinically advanced early-warning system available for this application.

Table 12
Event-based level comparison with the existing advanced methods.

Authors	Year	Dataset	Method	SEN (%)	FAR (/h)
Hung et al. (2010)	2010	Freiburg	wavelet and chaos theory	86.96	0.2500
Eftekhar et al. (2014)	2014		Threshold crossing	90.95	0.0600
Sharif and Jafari (2017)	2017		SVM	91.80–96.60	0.0500–0.0800
Yang et al. (2018)	2018		SVM	94.00	0.1100
Shokouh Alaei et al. (2019)	2019		Phase synchronization	100.00	0.1300
Zhang et al. (2020b)	2020		gradient boosting	91.67	0.1000
Wang et al. (2022a)	2023		1D-CNNs	98.65–100.00	0.0800–0.1200
Ours	2025		CANet	**100.00**	**0.0077**
Khan et al. (2017)	2018	CHB-MIT	CNN	87.80	0.1420
Affes et al. (2019)	2019		CGRNN	89.00	1.6000
Zhang et al. (2020a)	2020		CNN	92.90	0.1500
Yu et al. (2022)	2022		MLSTM	89.47	0.3400
Li et al. (2021)	2022		STS-HGCN-AL	95.50	0.1090
Gao et al. (2023)	2023		MSPPNet	93.80	0.0540
Zhang et al. (2023)	2023		STCNN	**98.13**	0.1500
Liu et al. (2024)	2024		P3DCNN-BiConvLstm3D-Attention3D	98.03	–
Ours	2025		CANet	97.06	**0.0251**

[1] SEN: the percentage of correctly predicted warnings among all seizure episodes; FAR: total false alarms produced by the epilepsy alert system per hour.
[2] Highlighted indicates the best-performing.

Future research will use diverse datasets to test the method's generalizability and predictive performance. We will investigate how channel count affects epilepsy prediction accuracy and whether fewer channels can still produce reliable results. We are currently focusing on epilepsy prediction research for patient-specific. In future work, we will conduct cross-patient epilepsy prediction research, try to develop an epilepsy prediction algorithm model that is applicable to cross-patients, and try to alleviate the problem of excessive differences in EEG signals between different patients.

5. Conclusion

We propose a method that employs a causal attention network combined with time, frequency, and channel feature fusion. We use dilated causal convolutions and standard convolutions to perform local feature extraction and fusion across the time, frequency, and channel domains. This allows the fused data to retain important information across these three dimensions. Subsequently, causal attention is employed to extract global causal features, further enabling the automatic extraction of more significant information. For the Freiburg intracranial EEG dataset, our method achieved 100.00% and 96.67% SEN with 0.0077/h and 0.0472/h FAR respectively for 1 h and 2 h durations of preictal periods. For the CHB-MIT scalp EEG dataset, it attained 97.06% and 92.31% SEN with 0.0251/h and 0.0666/h FAR respectively for 1 h and 2 h duration of preictal periods. The APT were 97.59 and 94.85 min for the two datasets, respectively. The method proposed in this study enhances the Sen, reduces the FAR, and improves the prediction timeliness compared to existing solutions by optimizing the algorithm model. Which is of great significance for improving the prognosis of patients.

CRediT authorship contribution statement

Yimin Qu: Writing – original draft, Formal analysis. **Songhui Rao:** Writing – original draft, Formal analysis. **Ting Li:** Formal analysis. **Ying Li:** Formal analysis. **Yan Niu:** Formal analysis. **Ruiyun Chang:** Formal analysis. **Xianchuan Chen:** Formal analysis. **Bin Wang:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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